Hovercraft Hockey Hackathon

Build a puck that moves by barely touching the floor.

Win condition: Best glide plus control in a gym-safe tournament.

THE SCIENCE YOU ARE USING

Air cushion cuts friction. A balloon pushes air down through a hole in a disc, lifting the disc on a thin cushion of air. With almost no contact, friction nearly vanishes, so a small push sends it gliding. Newton's laws then keep it moving until something stops it.

Glide versus control. Maximum glide is easy; control is hard. You balance lift and steerability to compete in gym-safe target hockey. Build quality and a clear explanation also score.

HOW TO BUILD AND RUN IT

01 **Mount the valve.** Glue a pop-top cap over the center hole of the disc, sealing the edge.

02 **Attach the balloon.** Stretch an inflated balloon over the closed cap so air escapes only through the hole.

03 **Test the glide.** Open the cap, set it down, and give a gentle push. Tune for a long, smooth glide.

04 **Practice control.** Learn to start and stop on target. Too much lift means no control.

05 **Play target hockey.** Compete in gym-safe rounds. Glide and control both count.

Safety: Latex allergy note. No face-level launches. Play floor-only. Allergy: Latex balloon alternative: small bags or school-approved nonlatex balloons.

MATERIALS

Recycled CDs or discs	per team
Pop-top bottle caps	per team
Balloons	per team
Hot glue	staff station
Targets	shared
Floor tape	shared

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ____________________

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The ONE variable we are testing: ____________________

Baseline result: ____________________

After redesign: ____________________

DEFEND YOUR DESIGN

What evidence made you change your design? ____________________

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If you ran it again, what is the ONE thing you would change? ____________________

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HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Glide performance	30
Target competition score	25
Explanation	20
Redesign	15
Build quality	10
Total	**100**